ortalama

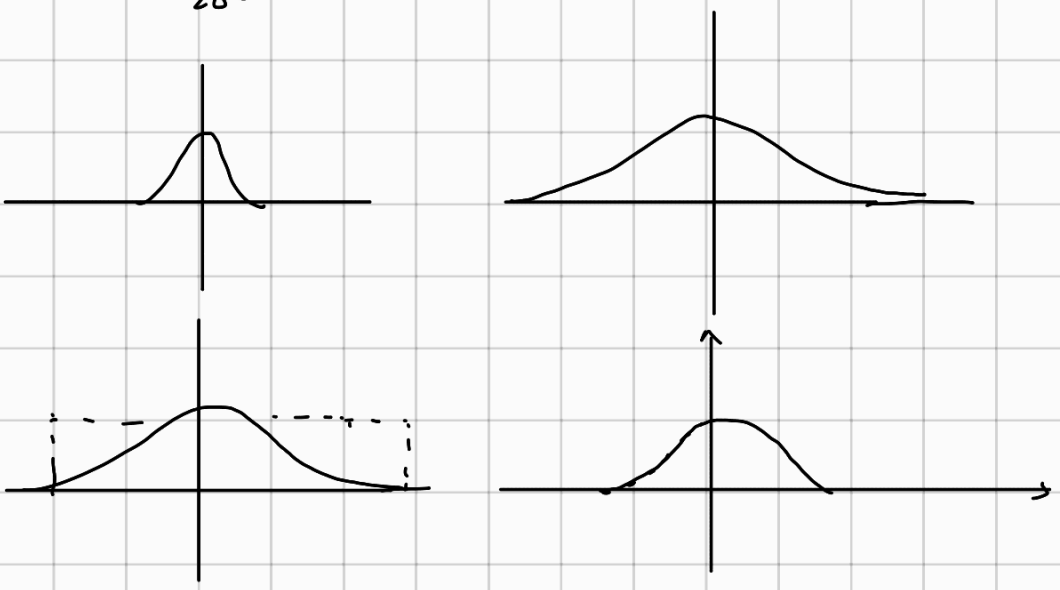
$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

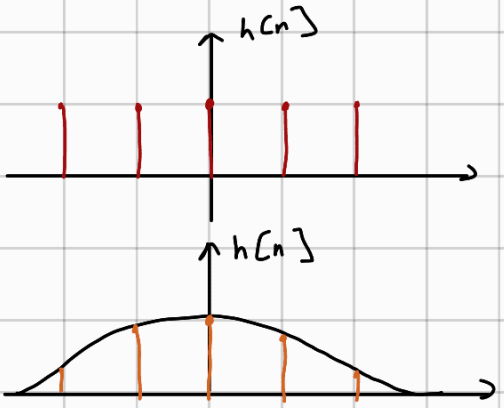
$\sigma\uparrow$ genişler

$$F(u) = \sqrt{\frac{\pi}{a}}\, e^{-\pi^2 u^2/a}$$

$\sigma\uparrow$ dumalır

$$a = \frac{1}{2\sigma^2}$$





$f * h \Leftrightarrow F \cdot H$

1	1	1
1	1	1
1	1	1

→ fourieri sinc

1	2	1
2	4	2
1	2	1

fourieri yauss

Laplacian:

$$\nabla^2 f(x,y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\nabla^2 f(x,y) \Leftrightarrow a \cdot -(u^2+v^2)\, F(u,v)$$

$u=0 \quad v=0 \Rightarrow 0$

$u=1 \quad v=1 \Rightarrow 2$

uzak noktalar çok hızlı artar

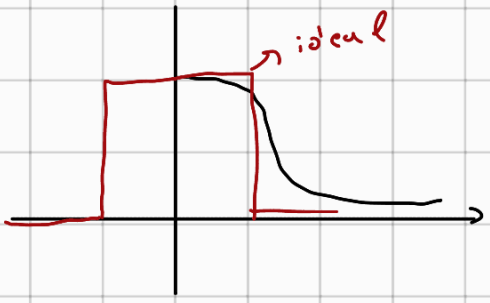
$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

$$G(u,v) = F(u,v) + \alpha(u^2+v^2)F(u,v)$$

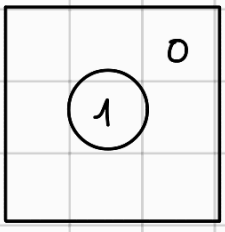
$$= F(u,v)\underbrace{[1+\alpha(u^2+v^2)]}_{H(u,v)}$$

↑ filtre

Bazen Frekansta daha hızlı yapılabiliyor işlemler

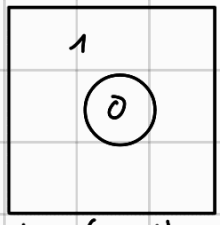


2 boyutlu fft shiftli



$H_{LP}(u,v)$



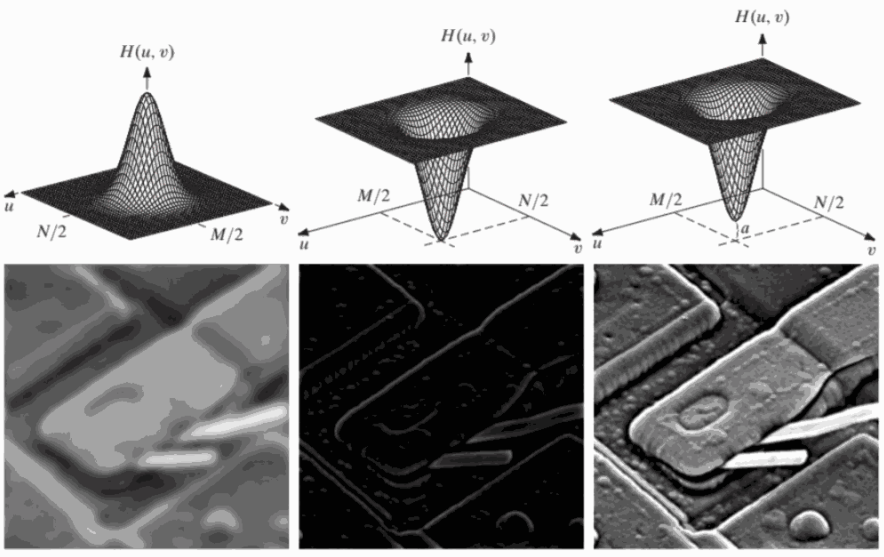


$H_{HP}(u,v) = 1 - H_{LP}(u,v)$



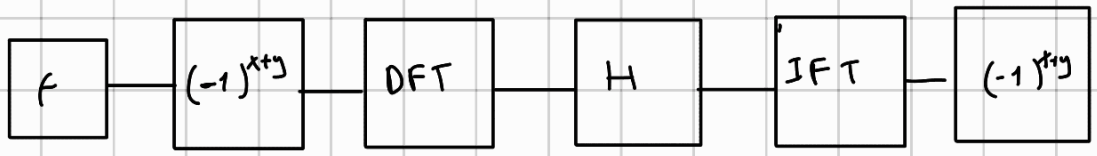


DC piksel değerleri ile alakalıdır



a b c
d e f

FIGURE 4.31 Top row: frequency domain filters. Bottom row: corresponding filtered images obtained using Eq. (4.7-1). We used $a = 0.85$ in (c) to obtain (f) (the height of the filter itself is 1). Compare (f) with Fig. 4.29(a).



H[k] k = 0, 1, 2

F[k] k = 0, ..., 99

a+b-1 = 102 noktalı fourier alınmalı

Zero padding

f (x,y) 100 x 100

F (u,v) 100 x 100

G(u,v) = F(u,v) H(u,v) ← ayrık ← DFT, DTFT de sürekli

g(x,y) = f(x,y) * h(x,y)

1	2	1
2	4	2
1	2	1

h(x,y) 3x3

yanlara 0 doldururuz

1- Çarpabilmek için boyutlar aynı olmalıdır H(u,v) 3x3

2- a+b-1 noktalı DFT alıp işlemleri yapıyoruz

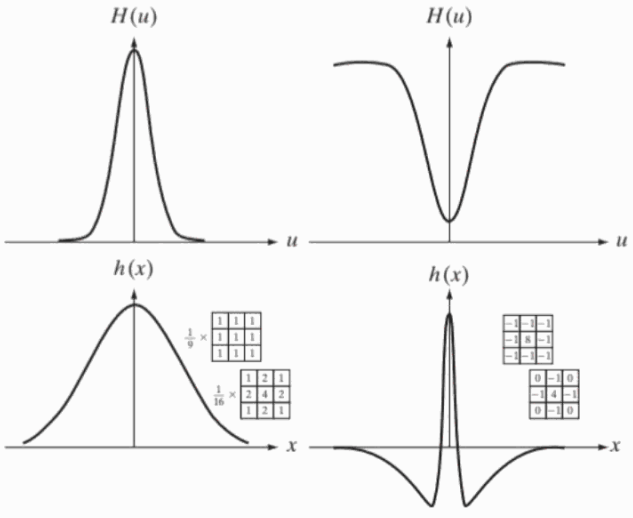
1	2	1
2	4	2
1	2	1

h(x, y) 3×3 f(x, y) 100×200

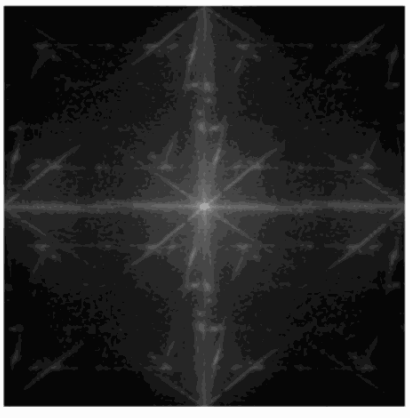
102 202

a c
b d

FIGURE 4.37
(a) A 1-D Gaussian lowpass filter in the frequency domain. (b) Spatial lowpass filter corresponding to (a). (c) Gaussian highpass filter in the frequency domain. (d) Spatial highpass filter corresponding to (c). The small 2-D masks shown are spatial filters we used in Chapter 3.







a b

FIGURE 4.38
(a) Image of a building, and (b) its spectrum.

Unsharp masking

$$g_{mask} = f(x,y) - \bar{f}_{LP}(x,y)$$

$$G_{mask} = F(u,v) - F(u,v)\, H_{LP}(u,v)$$

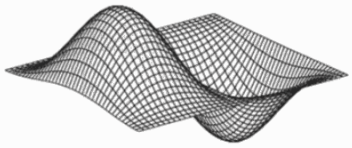
$$G_{mask} = F(u,v)\,\underbrace{(1 - H_{LP}(u,v))}_{H_{HP}(u,v)}$$

$$g(x,y) = f(x,y) + k\, g_{mask}(x,y)$$

$$G(u,v) = F(u,v)\,[1 + k\, H_p(u,v)]$$

a b
c d

−1	0	1
−2	0	2
−1	0	1



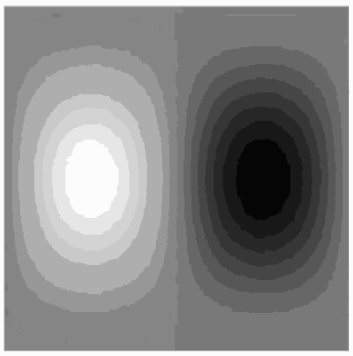
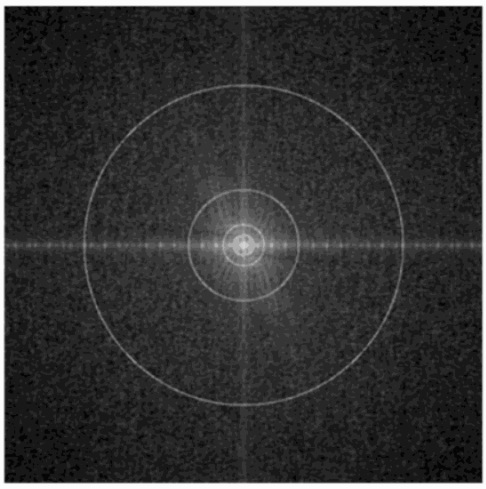






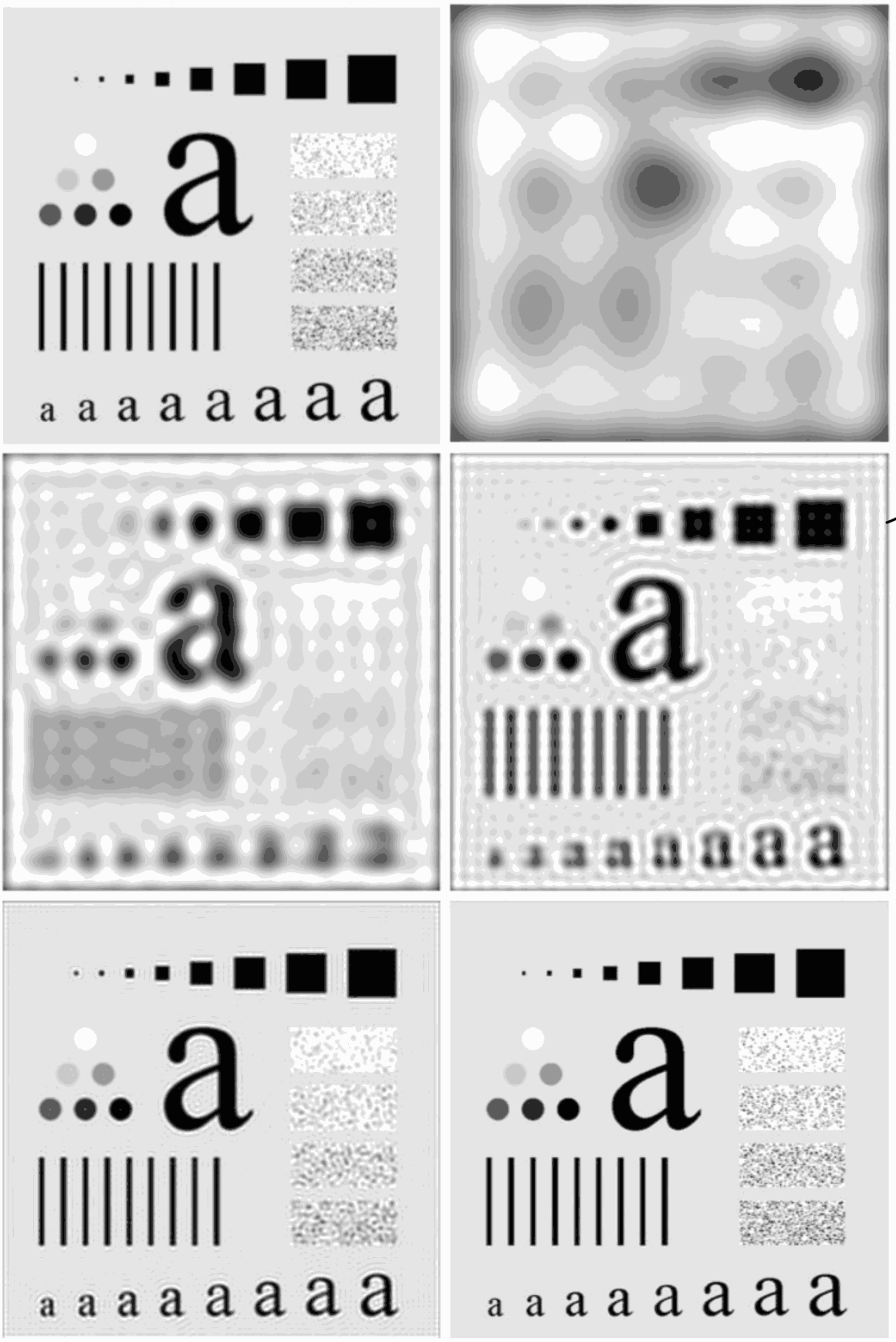
FIGURE 4.39 (a) A spatial mask and perspective plot of its corresponding frequency domain filter. (b) Filter shown as an image. (c) Result of filtering Fig. 4.38(a) in the frequency domain with the filter in (b). (d) Result of filtering the same image with the spatial filter in (a). The results are identical.





a b

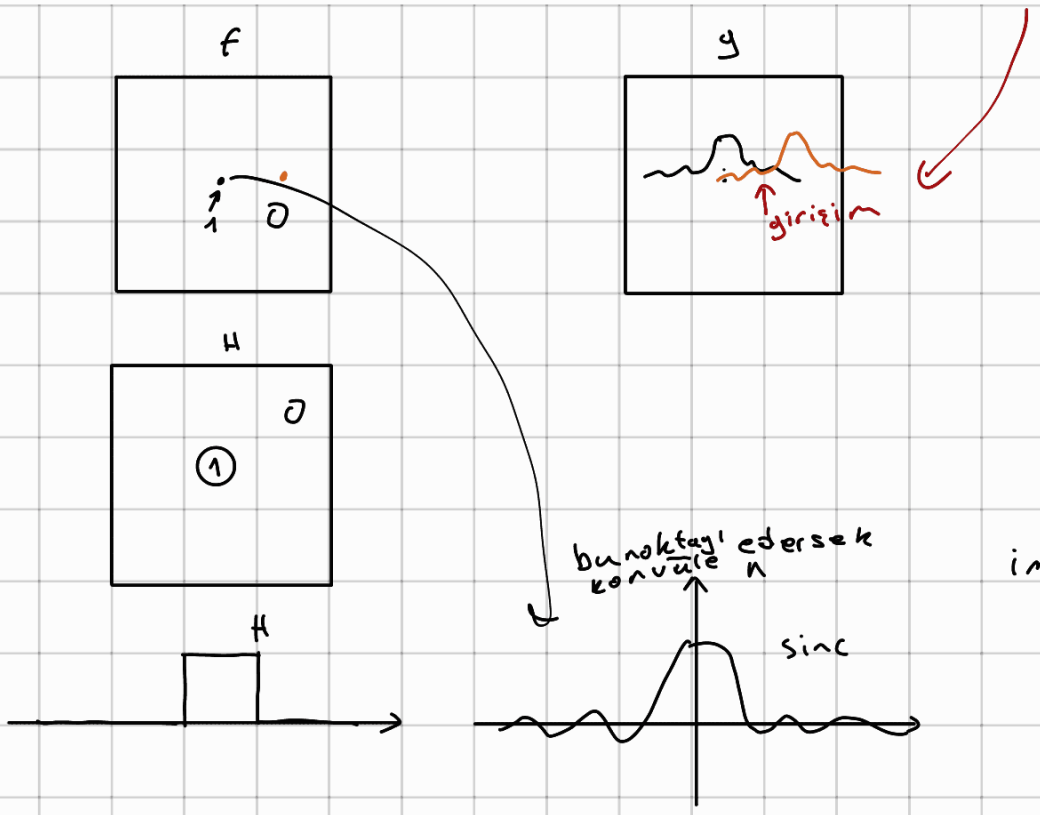
FIGURE 4.41 (a) Test pattern of size 688 × 688 pixels, and (b) its Fourier spectrum. The spectrum is double the image size due to padding but is shown in half size so that it fits in the page. The superimposed circles have radii equal to 10, 30, 60, 160, and 460 with respect to the full-size spectrum image. These radii enclose 87.0, 93.1, 95.7, 97.8, and 99.2% of the padded image power, respectively.



a b
c d
e f

FIGURE 4.42 (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.

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bunoktayı edersek konvole n

impulse alıyın için yine sinc'ler

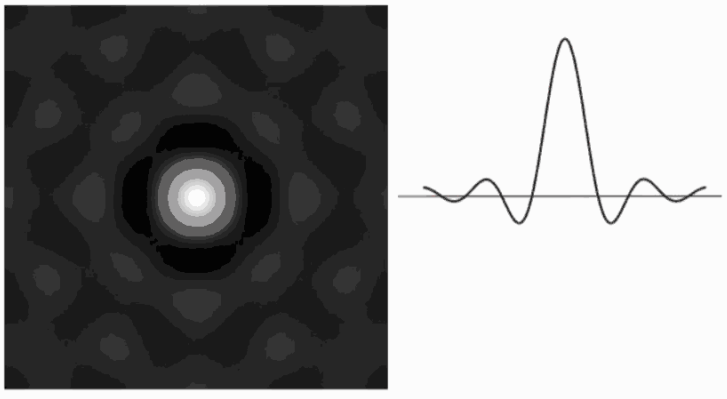
4.8.2 Butterworth Lowpass Filters

The transfer function of a Butterworth lowpass filter (BLPF) of order n, and with cutoff frequency at a distance D_0 from the origin, is defined as

$$H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}} \qquad (4.8\text{-}5)$$

where $D(u, v)$ is given by Eq. (4.8-2). Figure 4.44 shows a perspective plot, image display, and radial cross sections of the BLPF function.

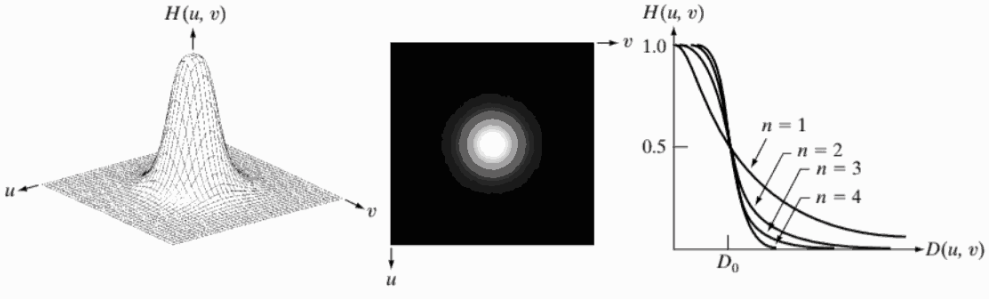
The transfer function of the Butterworth lowpass filter normally is written as the square root of our expression. However, our interest here is in the basic *form* of the filter, so we exclude the square root for computational convenience.



a b

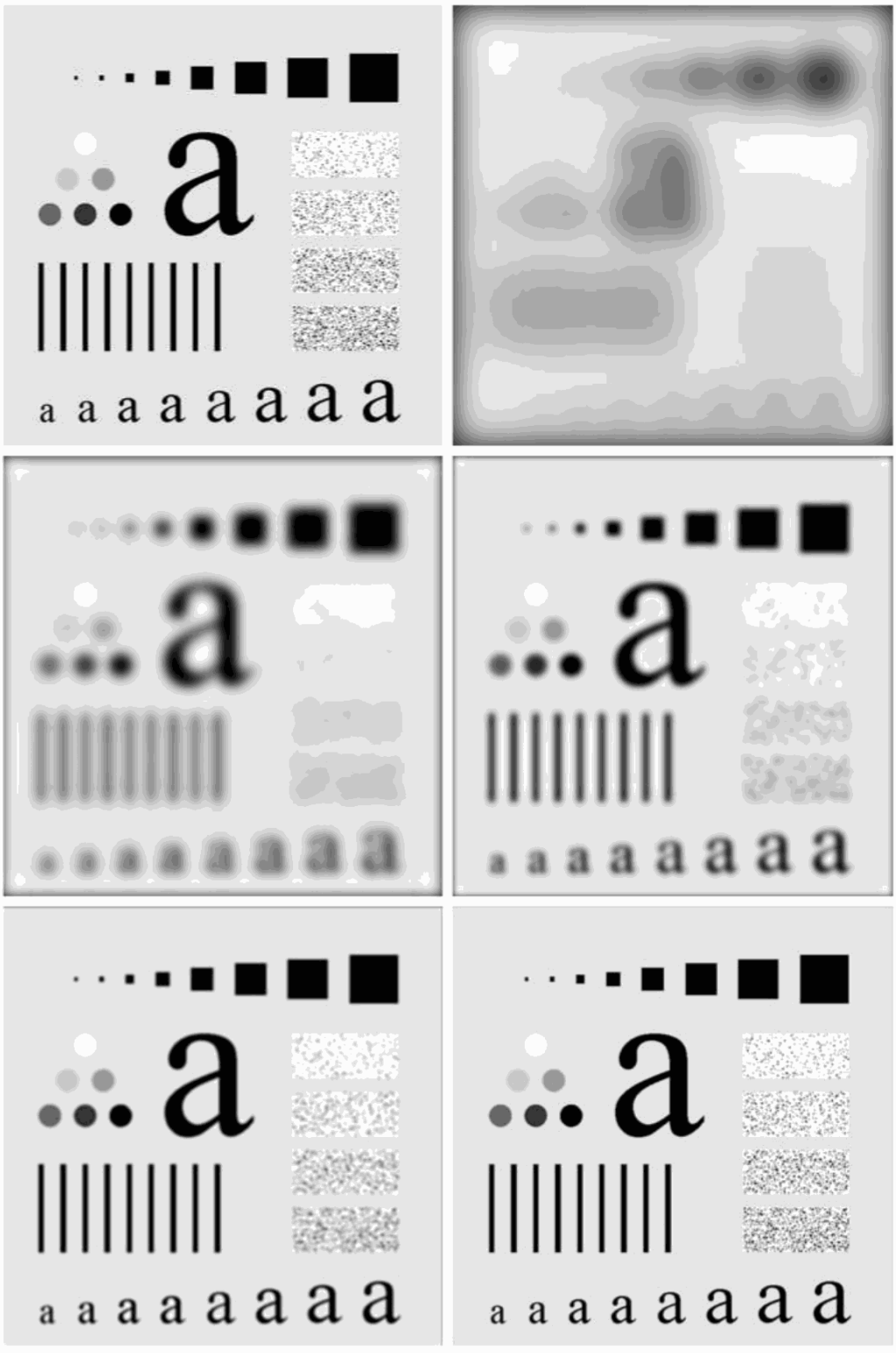
FIGURE 4.43 (a) Representation in the spatial domain of an ILPF of radius 5 and size 1000 × 1000. (b) Intensity profile of a horizontal line passing through the center of the image.

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a b c

FIGURE 4.44 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.



a b
c d
e f

FIGURE 4.45 (a) Original image. (b)–(f) Results of filtering using BLPFs of order 2, with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Fig. 4.42.

4.8.3 Gaussian Lowpass Filters

Gaussian lowpass filters (GLPFs) of one dimension were introduced in Section 4.7.4 as an aid in exploring some important relationships between the spatial and frequency domains. The form of these filters in two dimensions is given by

$$H(u, v) = e^{-D^2(u, v)/2\sigma^2} \tag{4.8-6}$$

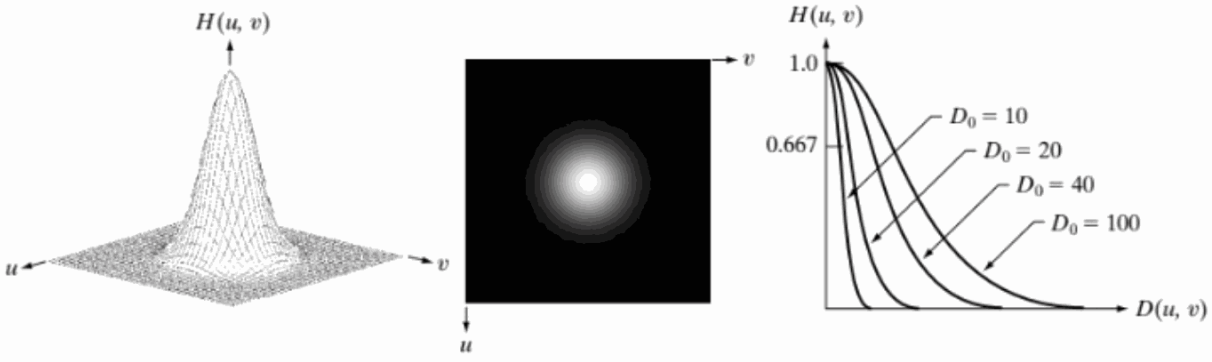
where, as in Eq. (4.8-2), $D(u, v)$ is the distance from the center of the frequency rectangle. Here we do not use a multiplying constant as in Section 4.7.4 in order to be consistent with the filters discussed in the present section, whose highest value is 1. As before, σ is a measure of spread about the center. By letting $\sigma = D_0$, we can express the filter using the notation of the other filters in this section:

$$H(u, v) = e^{-D^2(u, v)/2D_0^2} \tag{4.8-7}$$

where D_0 is the cutoff frequency. When $D(u, v) = D_0$, the GLPF is down to 0.607 of its maximum value.

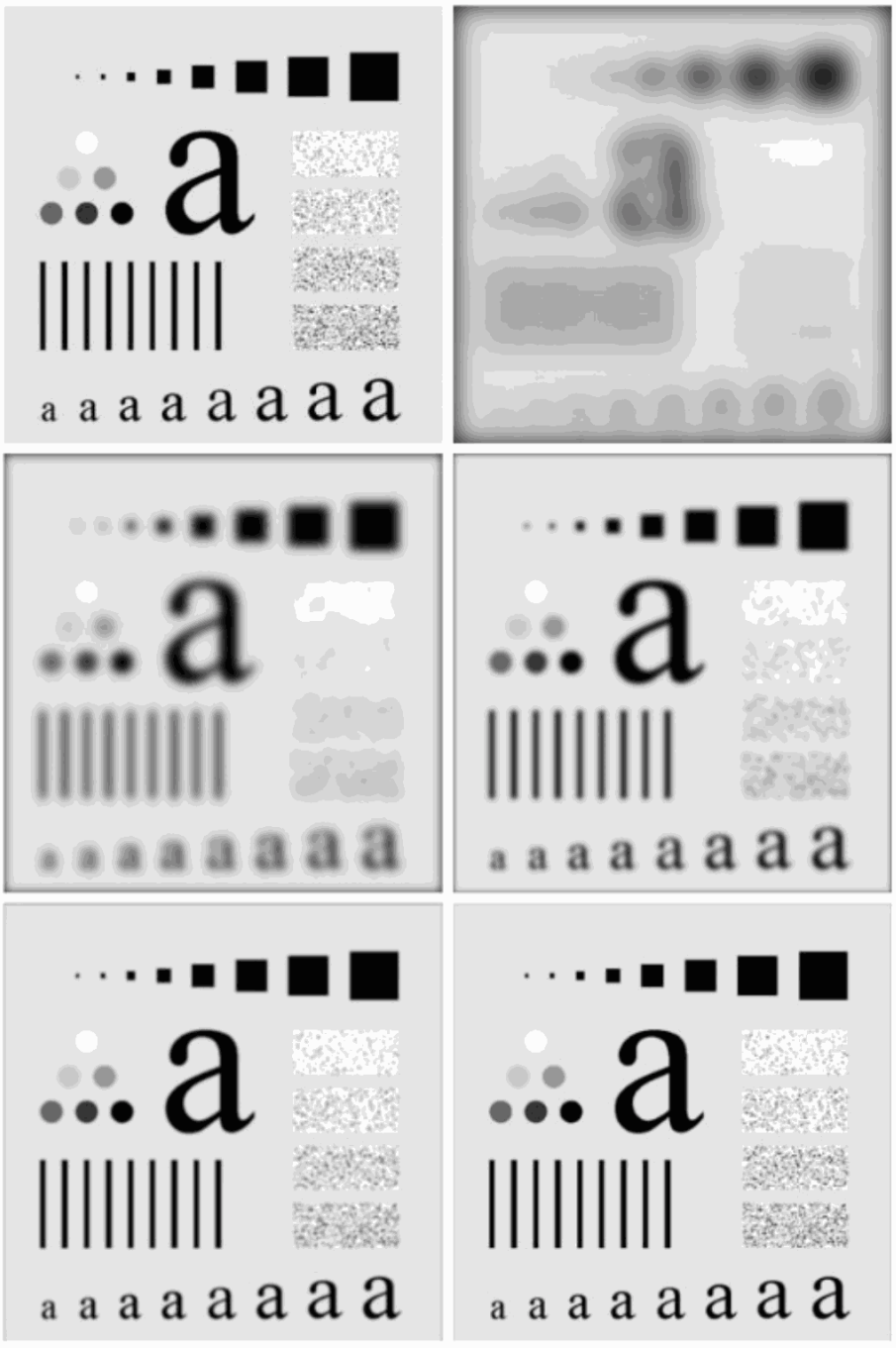
As Table 4.3 shows, the inverse Fourier transform of the GLPF is Gaussian also. This means that a spatial Gaussian filter, obtained by computing the IDFT of Eq. (4.8-6) or (4.8-7), will have no ringing. Figure 4.47 shows a perspective plot, image display, and radial cross sections of a GLPF function, and Table 4.4 summarizes the lowpass filters discussed in this section.

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a b c

FIGURE 4.47 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0.



a b
c d
e f

FIGURE 4.48 (a) Original image. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Figs. 4.42 and 4.45.



a b c

FIGURE 4.54 Results of highpass filtering the image in Fig. 4.41(a) using an IHPF with $D_0 = 30$, 60, and 160.

Unsharpmasking

$$g_{mask} = f(x,y) - f_{LP}(x,y)$$

$$G(u,v)_{mask} = F(u,v) - F(u,v) \cdot H_{LP}(u,v)$$
$$= F(u,v)\left[1 - H_{LP}(u,v)\right]$$
$$= F(u,v)\, H_{HP}(u,v)$$

$$g(x,y) = f(x,y) + k \cdot g_{mask}(x,y)$$
$$G(u,v) = F(u,v) + k\, F(u,v)\, H_{HP}(u,v)$$
$$= F(u,v)\left[1 + k H_{HP}(u,v)\right]$$

1 ↓ ofset var

keskinleştirme yaparken dc'yi korumalısın

Homomorphic Filtering

$f(x,y)$ = $i(x,y)$ · $r(x,y)$

$f(x,y)$: ışık miktarı ve ışık gansıma soğurma

$i(x,y)$: illumination

$r(x,y)$: reflection

$$\ln f(x,y) = \ln i(x,y) + \ln r(x,y)$$

$$F(\ln f(x,y)) = F(\ln(i(x,y)) + F(\ln r(x,y))$$

$$z(u,v) = F_i(u,v) + F_r(u,v)$$

yüksek frekans malzemeye karşılık geliyor demekki yüksek frekansları elemeyeceğiz.

(apply filter

$$z(u,v)H(u,v) = F_i(u,v)\cdot H(u,v) + F_r(u,v) H(u,v)$$

↓IFFT

↳ e^...

